

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Artificial Intelligence and Data Science

ASSESSMENT TOOL 2 – SCENARIO BASED

Course Code:	DSA03	Course Title:	Natural Language Processing
CO Assessed:	CO2 – Manipulation	Assessment:	Assessment Tool 2 – SCENARIO BASED
Weightage:	30% of CIA		
CO5 Statement:	Design inflectional, derivational, finite-state parsing, stemming algorithms and for analyse morphological methods. (BL-3)		
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Section / Batch:	B.Tech-AIDS/2024	Date:	03/08/2026

Code of Conduct

I S.Krishna Chaithanya(192424287) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: S .Krishna Chaithanya

Instructions

- Read all questions carefully before attempting the answers.
- Answer all five questions. Each question carries equal marks.
- Show all intermediate steps, calculations, probability computations, tagging decisions, and justifications wherever applicable.
- Duration: 30 minutes.

Section / Topic	Focus	Q Nos
English Word Classes, Tagsets for English, Part of Speech Tagging	Morphological decomposition of analyzing, analysis, and analytical; identification of roots, affixes, inflectional and derivational forms, and normalization for indexing.	Q1
English Word Classes, Tagsets for English, Stochastic Part of Speech Tagging	Morphological parsing of disagree, agreement, and agreeable; identification of prefixes, suffixes, base forms, and semantic effects of derivation.	Q2
Counting Words, English Word Classes, Part of Speech Tagging	Morphology-based normalization of govern, government, and governance for topic modeling, clustering, and lexical standardization.	Q3
Rule-Based Part of Speech Tagging, Transformation-Based Tagging	Morphological analysis of activate, activation, and reactivation; identification of derivational layers, word-class changes, and semantic shifts.	Q4
Rule-Based Part of Speech Tagging, English Word Classes, Part of Speech Tagging	Inflectional normalization of create, creates, and creating; extraction of root forms and classification of tense/aspect transformations.	Q5

Annexure A – SCENARIO BASED

1. A large-scale search engine indexing system processes user queries containing word variants such as “analyzing”, “analysis”, and “analytical”. The system must improve retrieval accuracy by grouping semantically related terms under a unified representation. Design a morphological processing approach to decompose each input word into root and affixes, distinguish between inflectional and derivational forms, and normalize them for indexing. Apply the process to the given inputs and determine the root form along with the morphological structure of each word.
2. An AI-based sentiment analysis platform receives product reviews containing words like “disagree”, “agreement”, and “agreeable”. The system must interpret sentiment consistently despite variations in word formation. Develop a morphological parsing strategy to identify prefixes, suffixes, and base forms, while capturing how derivational changes influence meaning. Process the given inputs to extract roots and classify each transformation, ensuring accurate semantic interpretation across all word forms.
3. A text analytics pipeline in a news aggregation system handles documents containing terms such as “govern”, “government”, and “governance”. The system must standardize these variations for topic modeling and clustering. Construct a morphology-based normalization mechanism that decomposes each word into its root and affixes, identifies derivational layers, and maps them to a common base form. Apply the approach to the input words and derive the normalized representation along with their morphological breakdown.
4. A document classification system processes input words such as “activate”, “activation”, and “reactivation” to improve semantic grouping and indexing.
Construct a morphological parsing strategy to identify prefixes, suffixes, and root forms while distinguishing derivational layers.
Analyze how prefix addition and suffix transformation affect meaning and word class.
Apply the process to decompose each word and map them to a unified base representation.
Derive the morphological structure and normalized root for each input word.
5. A search optimization module processes input words such as “create”, “creates”, and “creating” to ensure consistent retrieval across grammatical variations. Design a morphology-based normalization approach focusing on inflectional transformations such as tense and aspect.
Identify suffix patterns and map all variations to a common base form without altering meaning.
Apply the process to extract root forms and classify each transformation.
Determine the morphological breakdown and normalized output for each word.

1.Morphological Processing for Search Engine Indexing

Understanding of Scenario

A search engine receives different word forms such as analyzing, analysis, and analytical. Although they have different spellings, they share the same core meaning. Morphological processing decomposes words into their root (base form) and affixes (prefixes/suffixes), identifies whether the affixes are inflectional or derivational, and normalizes them to a common representation. This improves document retrieval by ensuring that all related words are indexed together.

Identification of Key Issues

- Different word forms reduce retrieval accuracy if indexed separately.
- Inflectional suffixes change grammatical form but not meaning.
- Derivational suffixes create new words or change the part of speech.
- A common root must be identified for efficient indexing.

Application of Concepts

Morphological Processing Steps

1. Tokenization – Read the input words.
2. Morphological Analysis – Separate the root and suffix.
3. Affix Classification – Identify whether the suffix is inflectional or derivational.
4. Normalization/Lemmatization – Convert all words to the common root analyze.
5. Indexing – Store the normalized form for efficient search.

Morphological Analysis Table

Input Word	Morphological Structure	Affix Type	Root Form (Normalized)
analyzing	analyze + -ing	Inflectional (Present participle)	analyze
analysis	analyze + -sis	Derivational (Forms a noun)	analyze
analytical	analysis + -tic + -al (or analyt + -ical)	Derivational (Forms an adjective)	analyze

Decision Making

- Remove the inflectional suffix -ing from analyzing to obtain analyze.
- Recognize analysis as a derivational noun related to analyze.
- Recognize analytical as a derivational adjective derived from the same word family.
- Normalize all three words to the common indexing term analyze.

Final Root Forms

Word	Final Root
analyzing	analyze
analysis	analyze
analytical	analyze

Conclusion

The morphological processing approach successfully decomposes each word into its root and affixes, identifies whether the affixes are inflectional or derivational, and normalizes all related forms to the common root analyze. By indexing the normalized root instead of separate word variants, the search engine improves retrieval accuracy, reduces redundancy, and ensures that documents containing semantically related terms such as analyzing, analysis, and analytical are retrieved together.

2.Morphological Parsing for AI-Based Sentiment Analysis

Understanding of Scenario

An AI-based sentiment analysis system processes product reviews containing words such as disagree, agreement, and agreeable. Although these words share the same base, prefixes and suffixes modify their meaning and grammatical category. Morphological parsing identifies the root, prefixes, and suffixes, classifies the transformations, and normalizes related forms to improve sentiment interpretation.

Identification of Key Issues

- Different word forms may express different sentiments.
- Prefixes can change the meaning (e.g., dis- indicates negation).
- Derivational suffixes change the grammatical category.
- The system must identify the common root while preserving semantic differences.

Application of Concepts

Morphological Parsing Steps

1. Tokenization – Read the input words.
2. Prefix Detection – Identify prefixes such as dis-.
3. Suffix Detection – Identify derivational suffixes like -ment and -able.
4. Root Extraction – Determine the common base form agree.
5. Semantic Interpretation – Preserve meaning changes caused by prefixes and suffixes.

Morphological Analysis Table

Input Word	Morphological Structure	Transformation Type	Root Form
disagree	dis- + agree	Prefix (Derivational – Negation)	agree
agreement	agree + -ment	Suffix (Derivational – Noun formation)	agree
agreeable	agree + -able	Suffix (Derivational – Adjective formation)	agree

Decision Making

- disagree: Detect the prefix dis-, which reverses the meaning to indicate negative sentiment.
- agreement: Recognize the suffix -ment, which converts the verb into a noun while preserving the positive concept.
- agreeable: Recognize the suffix -able, which forms an adjective describing a positive quality.
- Normalize all words to the root agree while retaining the semantic effect of prefixes and suffixes.

Conclusion

The morphological parsing strategy successfully identifies prefixes, suffixes, and the common root agree while distinguishing derivational transformations. By normalizing related words and preserving the meaning changes introduced by prefixes and suffixes, the sentiment analysis system achieves more accurate semantic interpretation and improves the consistency of sentiment classification across different word forms.

3.Morphology-Based Normalization for Text Analytics

Understanding of Scenario

A news aggregation system processes documents containing words such as govern, government, and governance. These words belong to the same word family but have different forms due to derivational suffixes. A morphology-based normalization mechanism decomposes each word into its root and affixes, identifies derivational layers, and maps them to a common base form for accurate topic modeling and document clustering.

Identification of Key Issues

- Word variations may be indexed as separate terms.
- Derivational suffixes create different grammatical forms.
- A common base form is needed for consistent topic modeling.
- The system must preserve semantic relationships while normalizing words.

Application of Concepts

Morphology-Based Normalization Steps

1. Tokenization – Read the input words.
2. Morphological Analysis – Separate the root and suffixes.
3. Derivational Layer Identification – Determine the derivational suffixes.
4. Root Extraction – Identify the common base form govern.
5. Normalization – Map all related words to the normalized root for indexing and clustering.

Morphological Analysis Table

Input Word	Morphological Structure	Derivational Layer	Normalized Root
govern	govern	Base verb	govern
government	govern + -ment	Derivational (Noun formation)	govern
governance	govern + -ance	Derivational (Abstract noun formation)	govern

Decision Making

- govern is identified as the base verb.
- government is formed by adding the derivational suffix -ment, creating a noun.
- governance is formed by adding the derivational suffix -ance, creating an abstract noun.
- Normalize all three words to the common root govern for indexing, clustering, and topic modeling.

Conclusion

The morphology-based normalization mechanism successfully decomposes each word into its root and derivational suffixes, identifies the derivational layers, and maps all related forms to the common base govern. This normalization improves topic modeling, document clustering, and information retrieval by ensuring that semantically related terms are treated as a single concept while preserving their morphological relationships.

4.Morphological Parsing for Document Classification

Understanding of Scenario

A document classification system processes words such as activate, activation, and reactivation. These words belong to the same morphological family but differ due to prefixes and suffixes. Morphological parsing identifies the prefixes, suffixes, and root, distinguishes derivational layers, and normalizes all related forms to a common base representation for improved semantic grouping and indexing.

Identification of Key Issues

- Word variations may reduce classification accuracy if treated separately.
- Prefixes modify the meaning of the base word.
- Derivational suffixes change the grammatical category.
- A unified base form is required for effective indexing and document classification.

Application of Concepts

Morphological Parsing Steps

1. Tokenization – Read the input words.
2. Prefix Detection – Identify prefixes such as re-.
3. Suffix Detection – Identify derivational suffixes such as -ation.
4. Root Extraction – Determine the common base form activate.
5. Normalization – Map all related words to the common root while preserving semantic changes.

Morphological Analysis Table

Input Word	Morphological Structure	Transformation Type	Normalized Root
activate	activate	Base verb	activate
activation	activate + -ion	Derivational (Verb → Noun)	activate
reactivation	re- + activate + -ion	Prefix (Repeat) + Derivational (Verb → Noun)	activate

Decision Making

- activate is identified as the base verb.
- activation adds the derivational suffix -ion, changing the verb into a noun.
- reactivation adds the prefix re-, meaning "again," and the suffix -ion, forming a noun that refers to the process of activating again.
- Normalize all three words to the common base activate for indexing and semantic grouping.

Conclusion

The morphological parsing strategy successfully identifies prefixes, suffixes, and the common root activate while distinguishing derivational layers. It captures how the prefix re- changes the meaning and how the suffix -ion changes the word class from verb to noun. By normalizing all related forms to the base activate, the document classification system improves semantic grouping, indexing, and retrieval accuracy.

5.Morphology-Based Normalization for Search Optimization

Understanding of Scenario

A search optimization module processes words such as create, creates, and creating. These words differ only in their grammatical forms (tense and aspect) but have the same meaning. A morphology-based normalization approach identifies inflectional suffixes, extracts the common root, and normalizes all variations to a single base form for consistent document retrieval.

Identification of Key Issues

- Inflectional variations may reduce search accuracy if indexed separately.
- Inflectional suffixes change grammatical form but not meaning.
- A common base form is required for efficient indexing and retrieval.
- Semantic meaning should remain unchanged after normalization.

Application of Concepts

Morphology-Based Normalization Steps

1. Tokenization – Read the input words.
2. Suffix Detection – Identify inflectional suffixes such as -s and -ing.
3. Morphological Analysis – Separate the root and suffix.
4. Root Extraction – Determine the common base form create.
5. Normalization – Map all word forms to the common root for indexing.

Morphological Analysis Table

Input Word	Morphological Structure	Transformation Type	Normalized Root
create	create	Base verb	create
creates	create + -s	Inflectional (Third-person singular present tense)	create
creating	create + -ing	Inflectional (Present participle/Continuous aspect)	create

Decision Making

- create is identified as the base verb.
- creates contains the inflectional suffix -s, indicating third-person singular present tense.
- creating contains the inflectional suffix -ing, indicating the present participle or continuous aspect.
- Normalize all forms to the common base create without changing their semantic meaning.

Conclusion

The morphology-based normalization approach successfully identifies inflectional suffixes, extracts the common root create, and maps all grammatical variations to a unified base form. Since inflectional transformations affect only tense and aspect without changing meaning, normalization improves search accuracy, indexing efficiency, and consistent retrieval of semantically identical terms.

Rubrics for Calculation

Criteria	Weightage (%)	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Understanding of Scenario	20%	Demonstrates clear and complete understanding of the scenario and context	Shows good understanding with minor gaps	Partial understanding; misses key aspects	Misinterprets or fails to understand the scenario
Identification of Key Issues	20%	Accurately identifies all relevant issues and factors involved	Identifies most key issues with minor omissions	Identifies limited issues; some irrelevant points	Unable to identify key issues
Application of Concepts	25%	Applies appropriate concepts effectively to address the scenario	Applies concepts with minor errors	Limited or partially correct application	Unable to apply relevant concepts
Decision Making	20%	Makes well-reasoned, logical decisions with strong rationale	Makes reasonable decisions with some justification	Makes weak decisions with limited justification	No clear decisions made or rationale provided
Clarity of Response	15%	Response is clear, structured, and logically presented	Generally clear with minor issues	Somewhat unclear or poorly structured	Disorganized and difficult to understand

Q. No. / Criteria Level	Understanding of Scenario	Identification of Key Issues	Application of Concepts	Decision Making	Clarity of Response	Sub. Total	Total %
1							
2							
3							
4							
5							

Faculty In-charge	Course Coordinator	Program Director
Signature: _____	Signature: _____	Signature: _____
Date: _____	Date: _____	Date: _____